

Answers: SAT vs ACT

SAT scores have a mean of 1500 and a standard deviation of 240.

ACT scores have a mean of 21 and a standard deviation of 6.

The z-score of a value is how many standard deviations it is above or below the mean:

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

a) $1740 - 1500 = 240$ points above the SAT mean.

b) $30 - 21 = 9$ points above the ACT mean.

c) Tim: $z = \frac{1740 - 1500}{240} = \frac{240}{240} = 1$

d) Sally: $z = \frac{30 - 21}{6} = \frac{9}{6} = 1.5$

e) **Sally** had the higher z-score (1.5 vs 1).

f) **Sally did better.** Her score is 1.5 standard deviations above the mean, and Tim's is only 1 standard deviation above. We cannot compare 240 points to 9 points directly because the tests use different scales. The z-scores put them on the same scale.

g) A negative z-score means the value is **below the mean**. For example, $z = -2$ means 2 standard deviations below the mean.